ADAPTIVE MANAGER AGENT

Member 5: System Orchestrator & Personal Adaptive Model Engine

AGENT 5 OF 5 — ORCHESTRATOR

1. AGENT OVERVIEW & CORE PURPOSE

The **Adaptive Manager Agent** is the central orchestrator and the primary innovation (USP) of the project. It does not perform direct tutoring or schedule creation. Instead, it maintains the user's **Personal Adaptive Model**, monitors behavioral deviations across all modules, and coordinates agent workflows to adjust overall system strategy.

Core USP Responsibility

Continuously updates behavioral profiles (attention span, procrastination triggers, mastery velocity) to predict friction before failure occurs.

System Orchestrator

Routes control signals across Tutor, Assessment, Planning, and Coaching agents in a closed feedback loop.

2. UNIVERSAL SYSTEM DATA SCHEMA OWNERSHIP

The Adaptive Manager owns and updates the `adaptive_model` state based on feedback from all sub-agents:


```
{ "adaptive_model": { "effective_focus_duration_mins": 22, "procrastination_patterns": ["Heavy lecture days", "Complex Math topics"], "task_delay_factor": 1.4, "learning_rate_velocity": 0.68, "current_system_mode": "PREEMPTIVE_RESTRUCTURE" } }
```


3. INPUT & OUTPUT SPECIFICATION

Type	Input Sources / Formats	Outputs / Artifacts Produced
Data Payload	• Execution logs from Planning Agent • Diagnostic test results from Assessment Agent • Reflection metrics from Life Coach Agent	• Updated `adaptive_model` JSON updates • Global system state events & orchestration routes • Preemptive intervention commands
Control Signals	• Discrepancy triggers (e.g., failed quiz, missed schedule block) from any sub-agent	• Explicit dispatch signals instructing sub-agents to take action (e.g., `DISPATCH_TUTOR`, `RESTRUCTURE_PLAN`)

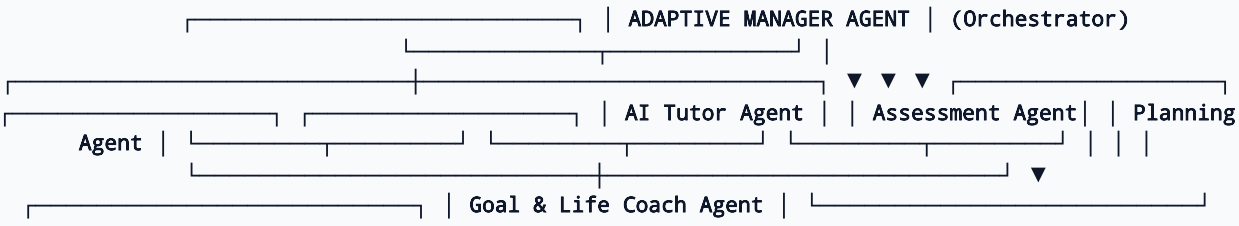
4. CLOSED LOOP ORCHESTRATION WORKFLOW

The Adaptive Manager executes an engine loop based on the **Observe → Understand → Predict → Adapt → Execute → Measure** paradigm:

1. **Observe:** Ingests events from all four sub-agents into a unified timeline.
2. **Predict Problems:** Evaluates active workload against historical procrastination patterns to flag imminent bottlenecks.
3. **Adapt Strategy:** Modifies parameters in `adaptive_model` (e.g., reducing study block size from 45 min to 20 min).
4. **Orchestrate:** Dispatches target agents with specific operational contexts.

5. ORCHESTRATION ARCHITECTURE & MULTI-AGENT LOOP

The diagram below illustrates how the Adaptive Manager coordinates all four sub-agents into a single unified system:



6. DETAILED AGENT ROUTING & COORDINATION MATRIX

Target Sub-Agent	Control Command Dispatched	Trigger Condition
Assessment Agent	`GENERATE_DIAGNOSTIC_EXAM`	Tutor Agent reports completed topic or mastery drops below threshold.
AI Tutor Agent	`INITIATE_REMEDIAL_SESSION`	Assessment Agent reports specific concept failure tag.
Personal Planning Agent	`PREEMPTIVE_SCHEDULE_RESTRUCTURE`	Manager predicts schedule failure based on upcoming lecture density.
Goal & Life Coach Agent	`TRIGGER_ACCOUNTABILITY_CHECKIN`	Procrastination risk exceeds alert thresholds for 2 consecutive days.

7. VALUE PROPOSITION & INNOVATION CONTRIBUTION

The Adaptive Manager embodies the core innovation of the graduation project:

- **True Personalization Engine:** Replaces static schedule tools with a system that continuously learns how each student actually works over time.
- **Preemptive Friction Prevention:** Solves bottlenecks before they occur by identifying behavioral patterns rather than reacting after a plan fails.
- **Multi-Agent Cohesion:** Converts individual AI capabilities into a synchronized, autonomous advisory system.